

SUKALYAN ROY

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PROFESSIONAL SUMMARY

Backend Engineer building data-intensive and ML-backed systems with **Python, FastAPI, PostgreSQL, Redis, and Azure**. Experience spans time-series EEG infrastructure, model-serving contracts, local-first retrieval, and deterministic AI-assisted workflows.

PROFESSIONAL EXPERIENCE

Backend Developer | Avinya Neurotech | Fully Remote | Jun 2025–Present

- Reworked a **five-service FastAPI platform**, exposing 50+ REST endpoints for EEG ingestion, analysis, export, and visualization workflows.
- Designed a **TimescaleDB schema**, using batched ingestion and time-windowed queries to support live and historical visualization of multichannel EEG samples.
- Streamed time-windowed database results in bounded WebSocket frames and cancelled stale requests; **profiling reduced first-chunk latency to 783 ms p50 and server send time to about 12 ms**.
- Implemented distributed task processing with Redis-backed job queues and Azure Blob Storage, supporting asynchronous exports and compute-intensive frequency-band analysis.
- Developed a standalone, contract-compatible PyTorch spike-detection framework for multichannel batch inference and interval export; production integration remains separate.
- Contributed bandpower, aEEG/qEEG, filter-parity, and Linux support to a cross-platform EEG viewer with real-time visualization and artifact-detection pipelines.
- Deployed and operated containerized services on **Azure**, managing **CI/CD releases**, service health monitoring, and production observability with **Prometheus and Grafana**.
- Maintained FDA DHF/DMR records and EU MDR technical documentation alongside explicit database migrations in a remote engineering workflow.

Tech Development Intern | Aecho AI | Fully Remote | Nov 2024–Jan 2025

- Built a local-first RAG assistant over 300+ pages using Python, Streamlit, and LM Studio to reduce retrieval time from minutes to seconds while keeping sensitive documents and inference off hosted APIs.

SELECTED PROJECTS & PUBLICATIONS

MediPredict ML Microservice & Research Manuscript | Dec 2025–Present | **FastAPI, scikit-learn, SHAP, Docker**

- Built a **FastAPI model-serving service**, with artifact-driven loading, typed contracts, health and readiness endpoints, Docker deployment, SHAP interpretability, and Spring Boot/React integration.
- Implemented reproducible preprocessing pipelines and **typed inference contracts**, preventing training-serving skew and artifact mismatches.
- Identified **target leakage** by separating quote-time inputs from renewal-time signals; used permutation importance to assess feature reliance and documented fairness and reproducibility limitations.
- Reduced the model artifact from 95.8 MB to 0.39 MB while lowering the train/test R^2 gap from 0.199 to 0.077; documented the system in an IEEE-format research manuscript.

Windowed EEG Signal-Processing Benchmark & Manuscript | May 2025–Aug 2025 | **Python, NumPy, SciPy, MNE**

- Built a **reproducible windowed EEG benchmark**, covering overlap-add, safe-window scheduling, edge policy, and bandpower equivalence on synthetic, CHB-MIT, and Sleep-EDF data.
- Measured worst-case p95 filter-plus-feature latency of **6.46 ms** on 8-channel CHB-MIT and **1.63 ms** on 2-channel Sleep-EDF, both within a 16.7 ms display-refresh budget.
- Documented failure modes including short-window zero-phase filtering, causal timing error, seam artifacts, and double-tapered spectral bias in an IEEE SPMB-format manuscript.

Tempo Priority Assistant | Jul 2026–Present | **Node.js, TypeScript, Python, Ollama**

- Engineered Tempo, a local-first **React/Express orchestrator** bridging **JSON-schema-constrained** Ollama generation with a deterministic Python service for accurate relative-date reconciliation.
- Implemented serialized inference queues for limited VRAM execution, **transforming unstructured task inputs into structured, browser-persistent schedules**.
- Added deterministic scheduling, capability boundaries, serialized local inference, and browser-local persistence with locked Node/Python test suites and CI.

Premiere Assist CLI | Jul 2026–Present | **Python, FFmpeg**

- Developed a local-first CLI producing canonical edit decision lists (EDL/XML) and analysis packages using **FFmpeg** for Adobe Premiere Pro integration.
- Engineered a **deterministic AI-operator interface**, restricting model outputs to validated candidate IDs and blocking arbitrary system commands.

Local-First RAG Chatbot with MCP Integration | Apr 2026–Present | **Python, FAISS, SQLite, Ollama**

- Engineered a **hybrid retrieval pipeline combining FAISS dense vector search with BM25 sparse search**, utilizing SQLite for inspectable metadata.
- On a fixed **18-query synthetic regression set**, hybrid retrieval retained Hit Rate@3 of 1.000 and improved MRR from **0.972 to 1.000**; results are regression evidence, not a general benchmark.

- Built a JSON-RPC Model Context Protocol (MCP) server for modular, structured access across Streamlit and command-line interfaces.

NewsTrellis Local News Evidence Organizer | Jul 2026–Present | Python, FastAPI, SQLite

- Built a local, single-user news evidence organizer with approval-gated RSS/Atom ingestion, SSRF-safe validation, retained storage, deduplication, search, event comparison, and deterministic digests.
- Verified the end-to-end workflow with **145 automated tests** and deterministic evaluation gates; broader usability testing and interface simplification remain.

TECHNICAL SKILLS

Primary: Python, SQL | **Working Familiarity:** JavaScript, TypeScript, PHP, C, Java
Backend and APIs: FastAPI, Node.js, Express, REST APIs, WebSockets, Distributed task processing
Databases and Data Systems: PostgreSQL, TimescaleDB, Redis, Azure Blob Storage, Vector DBs
ML, AI and Retrieval: PyTorch, scikit-learn, SHAP, RAG, Ollama, MCP, NumPy, pandas, FAISS
Cloud and DevOps: Azure, Docker, GitHub Actions, CI/CD, Kubernetes
Signal Processing: FFT, Spectrograms, Wavelet transforms, Bandpower analysis
Additional Tools: Linux/Bash, Prometheus/Grafana, OAuth

EDUCATION

Bachelor of Technology in Computer Science and Engineering

Netaji Subhash Engineering College | Kolkata | 2026

CGPA: 8.0/10.0

CERTIFICATIONS & ACHIEVEMENT

[Google Project Management Professional Certificate](#) | Harvard CS50: [CS50x](#), [AI with Python](#), [Cybersecurity](#)

Smart India Hackathon National Finalist, Government of India, 2023